

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained **Python-based backend services** using **Flask** and **FastAPI** frameworks, delivering **REST APIs** supporting customer-facing features across a high-traffic consumer platform; optimized request handling to reduce response latency by **35%** through async processing and efficient data serialization.
- Designed and managed **PostgreSQL relational database schemas** supporting reliable data storage for millions of daily transactions; wrote and optimized **SQL queries** improving query execution time by **42%** through indexing strategies, query analysis, and schema normalization across interconnected systems.
- Built **background processing jobs** and **automated workflows** using **Celery** and **APScheduler** to handle system tasks including data pipelines, scheduled reporting, and ETL operations; reduced manual data-movement overhead by **60%** and enabled real-time analytics ingestion.
- Supported deployment, monitoring, and reliability of **cloud-hosted systems** on **AWS** (EC2, S3, Lambda, CloudWatch); investigated and resolved production issues with **sub-1-hour MTTR** for P1 incidents; implemented **CloudWatch**-based alerting and automated remediation workflows reducing on-call escalations by **28%**.
- Developed **data pipelines** and **reporting workflows** using **Python** (pandas, NumPy) and **SQL** to consolidate and transform business data; created operational analytics dashboards and ad-hoc reporting outputs enabling stakeholders to extract actionable insights from data movement between systems and legacy platforms.
- Leveraged **AI-assisted development tools** and modern coding assistants to accelerate development cycles, automate testing workflows, and improve code quality; contributed to system improvements and modernization initiatives by identifying opportunities to refactor legacy components and simplify existing architecture through pragmatic problem-solving.

SKILLS

Backend & Python Development – Python, Flask, FastAPI, Django, REST APIs, asynchronous processing, Celery, APScheduler, background jobs.

Database & SQL – PostgreSQL, MySQL, Oracle, SQL query optimization, schema design, data modeling, database performance tuning, relational database management.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, CI/CD pipelines, version control, collaborative development practices.

Data Engineering & Analytics – Data pipelines, ETL workflows, pandas, NumPy, scikit-learn, data warehousing, reporting, analytics workflows, Power BI, Tableau.

Development Practices – Agile, Scrum, code reviews, change management, production support, incident management, monitoring, reliability engineering, AI-assisted coding tools.

Domain Knowledge – Enterprise business systems, payments operations, SWIFT messaging (MT/MX), ISO 20022, SEPA, cross-border payments, PSD2, Open Banking, regulatory compliance.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

High-Performance Data Analytics Backend Platform (HCL Technologies)

- Architected and deployed a **Python-based backend platform** using **FastAPI** and **PostgreSQL** to power a data-driven consumer platform supporting **2M+ daily active users**; designed **REST API endpoints** with comprehensive response caching and query optimization, reducing average response time from **800ms to 280ms (65% improvement)**.
- Built **automated data pipeline orchestration** using **Celery** and custom job schedulers to process incoming data streams, perform ETL transformations using **pandas** and **NumPy**, and load refined datasets into PostgreSQL; achieved **99.2% pipeline success rate** and reduced daily data ingestion time by **50%** to enable real-time reporting.
- Implemented comprehensive **database performance optimization** strategy including index design, query rewrites, and schema normalization across 12 interdependent services; reduced peak database CPU utilization by **38%** and improved transaction throughput by **45%** while maintaining ACID compliance.
- Deployed system monitoring and observability using **AWS CloudWatch**, structured logging, and custom dashboards to track API performance, background job health, and data pipeline reliability; configured automated alerting and incident response workflows reducing production downtime by **72%** year-over-year.
- Integrated **AI-assisted code analysis and testing tools** into development workflow to accelerate feature development and identify performance bottlenecks; leveraged AI-driven code suggestions to reduce feature delivery time by **33%** while maintaining code quality standards and backward compatibility.

SWIFT Payments Operations Dashboard & Real-Time Reporting System (HCL Technologies)

- Developed and maintained production-grade **Python backend services** and **SQL data pipelines** powering a SWIFT payments operations dashboard for a Tier-1 banking client; processed **500K+ daily payment messages** with **99.95% uptime SLA**, implementing real-time transaction-flow visibility and operational alerting for **ISO 20022** and **MT/MX message formats**.
- Built automated **data reconciliation workflows** and cross-border payment tracking APIs using **Python**, **PostgreSQL**, and **REST services** to support **SEPA instant payments** and PSD2 compliance; enabled operational teams to investigate payment discrepancies in **<5 minutes**, reducing manual reconciliation effort by **55%**.
- Implemented regulatory compliance and audit-trail infrastructure using **SQL** and application-layer controls to enforce **GDPR** data protection and financial-services governance; documented all state changes in immutable logs and designed automated data retention policies, ensuring 100% audit compliance across production systems.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)